Sachin Bala

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Technical Skills

SolidWorks (3D Design, FEA), AutoCAD, Minitab, JMP, LabVIEW, Six Sigma (Yellow Belt), Lean Manufacturing (5S, Kaizen), PFMEA, DFMEA, DOE, SPC, Process Capability Studies, Root Cause Analysis (8D, 5-Why), Python (Pandas, NumPy), SQL (MySQL), R, Data Visualization (Grafana, Matplotlib), Machine Learning, Computer Vision (YOLO), Cognex, Dr Schenk vision system, Microsoft Suite

Education

Western Michigan University (WMU) **Overall GPA: 3.69**

Bachelor's in Mechanical Engineering with minor in Computer Science and Mathematics

Work Experiences

Associate Process Development Engineer, Tesla Inc., Austin, TX **02/25-present**

- Commissioned two electrode production lines achieving 98% yield -- leading end-to-end equipment qualification, defining process specifications, and validating production readiness through structured capability studies and qualification protocols.
- Drove production optimization using Lean and Six Sigma methodologies (FMEA, DOE, SPC, 8D, 5-Why, Capability Study) to improve manufacturability, increase throughput, and reduce downtime across high-volume production equipment.
- Developed data collection and analysis pipelines using Python, R, MySQL, and JMP to generate predictive insights into equipment performance; collaborated with cross-functional engineering and operations teams to execute process improvements.

Engineering Technician, Process Development Team, Tesla Inc., Austin, TX **8/23-02/25**

- Qualified production equipment to meet stringent product specifications; applied statistical capability studies and process experiments to achieve a 10% increase in tool production rate while sustaining quality and cell performance targets.
- Designed and deployed real-time equipment monitoring dashboards in Grafana; authored SOPs for sensor integration and equipment usage to ensure consistent and reliable operation across the production floor.
- Conducted structured root cause analysis on equipment downtime events using SPC charts, fault logs, and process historians; implemented corrective actions that reduced unplanned stoppages and improved equipment uptime and reliability.

Engineering Co-Op, Scherdel SST, Muskegon, MI **8/22-12/22**

- Researched manufacturing material trends and analyzed EV component data to support product launch decisions and quality process design; collaborated with cross-functional teams across engineering, operations, and supply chain.

Process Engineering Intern, Scherdel SST, Muskegon, MI **05/22-08/22**

- Analyzed spring manufacturing process to detect failure modes and built a predictive ML model, improving production efficiency and accuracy by 25%; validated using Gauge R&R and F1-score methodology.
- Developed an automated Computer Vision defect detection system for production line quality inspection; validated system reliability using structured quantitative methods (Gauge R&R, F1-score analysis).

Laser Tech Technician Internship, Technique Inc., Jackson, MI **09/21-04/22**

- Optimized laser cutting processes using CAD and machine troubleshooting to manufacture vehicle chassis parts; validated parts via design review and laser tracker measurement data.
- Updated SOPs and trained technicians and operators to follow standardized procedures.

Engineering Research and Project

Senior Design Project - Two Tier Industrial Cart (Technique Inc.) **01/22-12/22**

- Designed a two-tier industrial cart in SolidWorks integrating a scissor lift mechanism; validated structural integrity with FEA analysis and numerical calculations to ensure manufacturability and safety.
- Applied DFMEA to identify and mitigate design risk, establishing data-backed design controls and a quality control plan.

Honors and Membership

Diether H. Haenicke Scholarship, WMU Dean's List, Yellow Belt Training (Six Sigma), Root Nepal (Co-Founder)